

CheckVAERS

An on-device search interface
to the public VAERS COVID-19 dataset.

889,521

VAERS REPORTS

56

STATES & TERR.

2020–25

YEARS COVERED

5

DOSES / CHECK

OVERVIEW

In one sentence

CheckVAERS lets a patient check whether a report matching their **state**, **sex**, **age at vaccination**, and **dose date(s)** exists in HHS's public COVID-19 adverse-event dataset — without transmitting any of that information off their device.

Site	App (PWA)	Source
checkvaers-site.vercel.app	check-vaers.vercel.app	github.com/Gaffattack54/checkvaers-mobile

CONTEXT

Why it exists

The public VAERS dataset is released as ~700 MB of CSVs at vaers.hhs.gov/data/datasets.html. The patient who asks “*is my report in there?*” has no reasonable path through that workflow, and the official lookup tool (CDC WONDER) is desktop-only and not built for one-off self-queries.

CheckVAERS is the missing four-question interface over the same data: surface matching VAERS_IDs, or honestly say *no match* — *here are the partial matches to review*.

DATA

Provenance, coverage, and de-identification

Source	vaers.hhs.gov public CSV exports (VAERSDATA, VAERSVAX, VAERSSYMPTOMS)
Coverage	2020 – 2025
Filter	VAX_TYPE = "COVID19" with a valid VAX_DATE
Records	889,521
Manufacturer split	Pfizer/BioNTech 47% · Moderna 46% · Janssen 6% · Novavax <1% · Unknown <1%
De-identification	Performed upstream by HHS per HIPAA Privacy Rule §164.514(b) Safe Harbor before public release. CheckVAERS only ever consumes the already-de-identified version.
Refresh cadence	Manual re-prep against the latest published CSVs. Current snapshot tagged v0.1.1-data .
Snapshot size	15 MB gzipped (≈60 MB uncompressed), served same-origin via an edge proxy to satisfy browser CORS constraints on the upstream GitHub-hosted release.

ALGORITHM

Matching, in spec form

Input. state (USPS 2-letter), sex (M / F / U, matching VAERS schema), dob → ageYears, and vaccineDates[] (1–5 ISO YYYY-MM-DD dates).

BUCKET	RULES	NOTE
Exact	state and sex equal; ageDelta ≤ 1 year; any user dose date within ±7 days of the record's VAX_DATE.	Returned in full.
Potential	state and sex equal; ageDelta ≤ 5 years; any user dose date within ±30 days.	Capped at 20, sorted ascending by combined age + date delta. Exact-bucket records excluded.
None	Both buckets empty.	Result page surfaces the dataset metadata and offers the Report deep-link.

Records are stratified by a state-keyed index built once at data-load time. Per-check latency on the full 900k-record dataset is **under 10 ms on a phone CPU**, well below one frame.

PRIVACY

Why HIPAA does not attach

“We do not store PHI in the first place.”

The defensible answer to a reviewer asking how PHI is encrypted at rest.

- **No PHI is processed.** The dataset is de-identified by HHS before release; we read only that copy.
- **HIPAA does not attach.** CheckVAERS is a consumer-facing search tool, not a covered entity or business associate. The Privacy Rule does not apply to a consumer's voluntary self-query against public data.
- **User inputs never leave the device.** State, sex, DOB, and dose dates stay in the browser. Matching runs entirely in client-side JavaScript.
- **Local-only persistence.** Completed checks save to IndexedDB on the user's device. There is no server-side database.
- **HTTPS-only.** Snapshot is ETag-validated on first launch and cached for offline use.
- **No accounts. No tracking. No analytics. No third-party scripts.**

FLOW

Six steps, one screen apiece

1. **State** — searchable USPS dropdown.
2. **Sex** — M / F / Prefer not to say.
3. **Date of birth** — native date picker; age computed and displayed inline.
4. **Vaccine dose date(s)** — 1 to 5 entries.
5. **Review** — single screen, every row deep-linked back to its edit step.
6. **Result** — YES / MAYBE / NO headline with the VAERS_ID list, manufacturer breakdown, and a CDC WONDER deep link to the canonical full report.

Draft is autosaved to `sessionStorage` for refresh resilience. Completed checks appear in a per-device History tab keyed by the device, not by identity.

TRADE-OFF

A deliberate omission, called out plainly

The prepared snapshot omits four fields that would otherwise inflate the in-browser object graph past iOS Safari's per-tab memory budget: SYMPTOM_TEXT (narrative), RECVDATE, NUMDAYS, and the 4th–5th SYMPTOM codes.

Each result card shows: VAERS_ID, manufacturer, vaccination date, state, age, top 3 coded symptoms, and a **“View full report on CDC WONDER”** link by VAERS_ID. The full record — narrative, days to onset, treatment, recovery — is one click away at the canonical source.

This was an engineering choice, not a content choice: keep the browser memory budget safe on a six-year-old iPhone, and let CDC WONDER serve as the authoritative detail layer.

BOUNDARIES

What CheckVAERS is not

“A VAERS match is a report, not a diagnosis.”

The same framing CDC uses when describing VAERS' role in safety surveillance.

- Not medical advice. Not a diagnostic tool.
- Not a causation tool.
- Not a substitute for filing a VAERS report — the Report tab deep-links to vaers.hhs.gov/reportevent.html.
- Not affiliated with CDC, FDA, HHS, or any government agency.
- Not a real-time mirror of the live VAERS database. The snapshot lags.

ENGINEERING

Architecture in one paragraph

Next.js 14 (App Router, TypeScript), deployed as two Vercel projects from the same monorepo: a *site* build (rich desktop-first marketing landing) and an *app* build (mobile-first PWA, installable to home screen). The two share 100% of the functional code; a single `NEXT_PUBLIC_VARIANT` env var selects the home-route presentation. The client matcher is an in-browser state-indexed scan. Persistence is Dexie over IndexedDB (check history + data cache). The dataset is prepared by a Node script from raw HHS year CSVs and published as a GitHub Release asset; the client fetches it via a same-origin `/api/vaers-data` route that follows the upstream redirects server-side (the upstream `release-assets.githubusercontent.com` does not send CORS headers, so a direct fetch would fail). A service worker caches the app shell and the data file for offline use.

ROADMAP

What comes next

Near-term

- **Per-state lazy loading.** The API endpoint returns only the user's state slice on demand, dropping first-load payload from 15 MB to 1–3 MB and removing the mobile memory ceiling entirely.
- **Detail-on-demand.** A `/api/vaers-detail/[id]` route restores the narrative, days-to-onset, and receive-date fields per match — without re-bloating the in-browser dataset.

Medium-term

- Coverage beyond COVID-19 (flu, MMR, HPV, etc.) using the same prep pipeline and matcher.
- A VICP / CICP eligibility flowchart on the Learn tab.
- Optional opt-in Plausible analytics for usage metrics — privacy-respecting, no cross-site cookies.

Open question

- Whether a clinician-facing variant — full record fields client-side, desktop memory budget assumed — is worth a separate build.

REFERENCES

Links

Site (marketing)	https://checkvaers-site.vercel.app
App (PWA)	https://check-vaers.vercel.app
Source code	https://github.com/Gaffattack54/checkvaers-mobile

Current data snapshot <https://github.com/Gaffattack54/checkvaers-mobile/releases/tag/v0.1.1-data>

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